

Machine Learning - Mid-Semester 1 Shorts

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Session 1 - What Machine Learning Learns

[Know More Detailed External Sources](#)

Intuition

Machine learning uses examples to fit rules that are difficult to hand-code. Specify what the system does, what experience it receives, and how improvement is measured.

Definitions and jargon

Mitchell's framing uses task T , experience E , and performance measure P . Supervised uses labels; unsupervised seeks unlabeled structure; semi-supervised mixes labeled and unlabeled examples; reinforcement learning learns actions from rewards.

Formulas, symbols, shapes, and assumptions

For a supervised table, X has shape n by d : n examples and d features. Target y has n entries. Regression predicts a number; classification predicts a category or class probabilities. P must match the real cost: accuracy alone is often unsuitable when classes or error costs are unequal.

Conditional properties

A pattern generalizes only when experience represents the future population. Correlation does not establish causation; more irrelevant or poor-quality data need not help.

Important distinctions

Traditional programming supplies rules plus inputs to obtain outputs. Supervised learning supplies inputs plus known outputs to fit a rule. A model-based method fits a general parameterized rule; an instance-based method may retain examples and defer prediction work.

Checked example

For spam filtering, T is labeling new email, E is historical labeled email, and P could emphasize recall when missed spam is costly or precision when blocking legitimate mail is costly. "Get high accuracy" is incomplete until the positive class and error costs are defined.

Traps and counterexamples

k-nearest neighbours is instance-based supervised learning when labels are available; k-means is unsupervised clustering. "AI, ML, and deep learning are identical" is false: ML is one family of techniques used within the broader AI field, and deep learning is one ML family.

Answer-free recall

- 1 Give T , E , and P for demand forecasting.
- 2 Classify four scenarios by available feedback.
- 3 Explain how a performance measure changes the best model and how unrepresentative sampling breaks a deployment claim.

Session 2 - Data, Splits, Preprocessing, and Metrics

[Know More Detailed External Sources](#)

Intuition

A trustworthy workflow protects future-like data from fitting choices. Learned cleaning and scaling belong inside the training boundary.

Definitions and jargon

A feature is an input; a target is what is predicted. Leakage occurs when unavailable or held-out information influences training. Class imbalance means class frequencies differ greatly. A true positive (TP) is a correctly predicted positive, a false positive (FP) is a negative example incorrectly predicted positive, a true negative (TN) is a correctly predicted negative, and a false negative (FN) is a positive example incorrectly predicted negative.

Formulas, symbols, shapes, and assumptions

Standardize with training mean and standard deviation: $z = (x - \mu_{\text{train}}) / s_{\text{train}}$. Min-max scale with training extrema: $(x - \min_{\text{train}}) / (\max_{\text{train}} - \min_{\text{train}})$. Let $N = TP + TN + FP + FN$ be the total number of evaluated examples. Accuracy is $(TP + TN) / N$; precision is $TP / (TP + FP)$; recall is $TP / (TP + FN)$; F1 is $2(\text{precision})(\text{recall}) / (\text{precision} + \text{recall})$. A zero denominator makes the corresponding elementary ratio undefined.

Conditional properties

For a normal distribution, about 99.73% lies within $\mu \pm 3 \sigma$. Tukey fences $Q1 - 1.5 \text{ IQR}$ and $Q3 + 1.5 \text{ IQR}$ only flag observations. Standardization needs no Gaussian assumption. A constant training feature needs an explicit zero-denominator rule.

Important distinctions

Training fits parameters; validation chooses models, hyperparameters, and thresholds; test estimates frozen performance. Loss trains; a metric reports behavior. Stratification preserves class proportions but cannot repair biased collection.

Checked example

With $TP=12$, $FP=3$, $FN=8$, $TN=77$, accuracy is $89/100=0.89$, precision is $12/15=0.80$, and recall is $12/20=0.60$. Accuracy looks strong even though 40% of actual positives were missed.

Traps and counterexamples

Split before fitting scaling or imputation. An outlier flag does not prove corruption. For time-ordered prediction, a random split may leak the future.

Answer-free recall

- 1 Describe a leakage-safe split-and-scale sequence.
- 2 Calculate all four metrics from a confusion table.
- 3 State the conditions behind three-sigma and IQR rules.

Session 3 - Linear Regression and Bias-Variance

[Know More Detailed External Sources](#)

Intuition

Linear regression fits a weighted sum to a numeric target. Basis functions bend the input representation while the model remains linear in its coefficients.

Definitions and jargon

$\hat{y}$ is a prediction and residual $e = \hat{y} - y$. Least squares minimizes squared residuals. A basis function transforms inputs before weighting. Bias is systematic underfitting; variance is sensitivity to training-sample changes.

Formulas, symbols, shapes, and assumptions

For X of shape n by d , w is d by 1 , and $\hat{y} = Xw$ is n by 1 . Mean squared error is $MSE = (1/n) \|Xw - y\|^2$; its gradient is $(2/n) X^T (Xw - y)$. Normal equations are $X^T X w = X^T y$. The inverse form $w = (X^T X)^{-1} X^T y$ requires full column rank and acceptable conditioning.

Conditional properties

For $R^2 = 1 - SSE/SST$, $SST = \sum (y_i - \bar{y})^2$ must be positive; a constant target makes the ratio undefined. L2 ridge shrinks squared coefficients. L1 lasso is nonsmooth at zero but optimizable.

Important distinctions

Direct and iterative methods target the same unregularized least-squares objective. Compute every batch-gradient component from the same old w . Polynomial regression is nonlinear in input but linear in coefficients.

Checked example

For $\hat{y} = 2 + 3x$, at $x = 4$ the prediction is 14. If $y = 15$, residual is -1 and squared error is 1. For one coefficient with gradient -4, old value 2, and learning rate 0.1, the update is $2 - 0.1(-4) = 2.4$.

Traps and counterexamples

Explicit inversion is unsafe for dependent or nearly dependent columns. Low training error can mean overfitting. Large learning rates can overshoot; tiny ones converge slowly.

Answer-free recall

- 1 Give all shapes in the matrix model.
- 2 Derive the normal equations from the gradient condition.
- 3 Contrast high bias and high variance from training/validation errors.

Session 4 - Linear Classification and Decision Theory

[Know More Detailed External Sources](#)

Intuition

A linear discriminant divides feature space with a flat boundary; decision theory chooses the action with smallest expected cost.

Definitions and jargon

A discriminant $g_k(x)$ is a class score; choose the largest. A posterior $P(C_k|x)$ is probability after observing x . A generative classifier models priors and class-conditional inputs; a discriminative classifier models posteriors or boundaries.

Formulas, symbols, shapes, and assumptions

For binary linear score $g(x) = w^T x + b$, w and x are d by 1 ; boundary points satisfy $g(x) = 0$. The signed geometric distance is $g(x)/\|w\|_2$ only when w is nonzero. With loss $\lambda(a_i|C_j)$, conditional risk is $R(a_i|x) = \sum_j \lambda(a_i|C_j)P(C_j|x)$; choose the action of minimum risk.

Conditional properties

Equal error costs yield maximum-posterior classification; unequal costs shift the rule. Positive rescaling of w, b preserves boundary and signs but changes score magnitude.

Important distinctions

A score is not automatically a probability, distance, or calibrated confidence. Maximum likelihood estimates parameters from observed data; Bayesian decision theory combines probabilities with action costs.

Checked example

Suppose $P(\text{disease}|x) = 0.20$. Treating a healthy person costs 1; missing disease costs 10; correct actions cost 0. Risk(treat) is $1 \cdot 0.80 = 0.80$; risk(do not treat) is $10 \cdot 0.20 = 2.00$. Minimum-risk action is treat despite posterior below 0.5.

Traps and counterexamples

$g(x) = 8$ is not distance 8 without normalization. High false-negative cost can make a low-posterior action optimal. Linear boundaries need transformations for nonlinear patterns.

Answer-free recall

- 1 State the argmax discriminant rule.
- 2 Calculate two conditional risks and choose an action.
- 3 Explain why raw score magnitude is scale-dependent.

Session 5 - Logistic Regression and Multiclass Softmax

[Know More Detailed External Sources](#)

Intuition

Logistic regression turns a linear score into a number between zero and one. Softmax extends the idea to mutually exclusive classes whose probabilities sum to one.

Definitions and jargon

The sigmoid is $\sigma(z) = 1/(1+\exp(-z))$. Odds are $p/(1-p)$ and log-odds are their natural logarithm. A logit is an unnormalized class score. Cross-entropy penalizes probability assigned away from the true class.

Formulas, symbols, shapes, and assumptions

Binary: $z = w^T x + b$, $p = \sigma(z)$, $\ln(p/(1-p)) = z$, and loss $-[y \ln p + (1-y) \ln (1-p)]$. For K exclusive classes, $p_k = \exp(z_k) / \sum_j \exp(z_j)$ and one-example loss is $-\sum_k y_k \ln p_k = -\ln p_t$ for true class t .

Conditional properties

With a linear predictor, binary/multinomial cross-entropy and a convex L1 or L2 penalty form a convex objective; uniqueness can still depend on data and regularization. This does not make a neural-network objective convex. Numerically, subtract the largest logit before exponentiating softmax.

Important distinctions

Changing the classification threshold changes decisions but does not refit probabilities. One-vs-rest trains binary decisions separately; multinomial softmax fits a joint normalized model. Probability output need not be calibrated merely because it lies in $[0, 1]$.

Checked example

For logits $(0, \ln 2, \ln 3)$, exponentials are $(1, 2, 3)$ and sum to 6, so probabilities are $(1/6, 1/3, 1/2)$. If the true class is third, loss is $-\ln(1/2) = \ln 2$, about 0.693.

Traps and counterexamples

Avoid $\ln(0)$ by stable computation. Complete separation can send unregularized coefficient magnitudes very high. A coefficient expresses conditional association in log-odds, not causation.

Answer-free recall

- 1 Convert between score, probability, odds, and log-odds.
- 2 Compute a three-class softmax and cross-entropy.
- 3 State the convexity scope, why it does not extend to a multilayer neural network, and why threshold selection differs from probability fitting.

Session 6 - Decision Trees, Impurity, and Pruning

Guide Practice Sources

Intuition

A decision tree asks one feature question at a time. Training chooses splits that make child nodes purer; pruning trades a little training fit for a simpler model that may generalize better.

Definitions and jargon

A node contains examples, a split routes them to children, and a leaf predicts. Entropy and Gini impurity measure class mixing. Information gain is impurity removed by a split. MDL prefers the shortest combined description of model and unexplained data under a stated code.

Formulas, symbols, shapes, and assumptions

For class proportions p_k , entropy is $H = -\sum_k p_k \log_2(p_k)$ with $0 \log 0 = 0$; Gini is $G = 1 - \sum_k p_k^2$. For children c , $\text{gain} = I(\text{parent}) - \sum_c (n_c/n) I(c)$. A continuous split can test $x_j \leq t$ versus $x_j > t$.

Conditional properties

Entropy and Gini are zero at a pure node; balanced binary proportions maximize them at 1 bit and 0.5, respectively. Different impurity criteria can choose different splits. Continuous features need not be discretized beforehand; the tree can search thresholds.

Important distinctions

Pre-pruning limits growth; post-pruning removes branches after growth. MDL is not “choose the fewest nodes”: it balances the code length of the tree and remaining errors. Missing values require an explicit learned routing, surrogate, probabilistic-routing, or imputation policy.

Checked example

A parent has four positive and four negative examples, so entropy is 1 bit. A split produces children (3+, 1-) and (1+, 3-); each has entropy about 0.811, and equal child sizes give gain $1 - 0.811 = 0.189$ bit.

Traps and counterexamples

Deepest is not necessarily best on unseen data. Missing does not automatically mean zero. A high-cardinality feature can create misleadingly attractive splits unless complexity and validation are controlled.

Answer-free recall

- 1 Compute entropy, Gini, and gain for a split.
- 2 Explain continuous and missing-value handling.
- 3 Contrast pre-pruning, post-pruning, and MDL.
- 4 State why a pure child alone does not prove generalization and how validation-guided pruning responds.

Session 7 - kNN, Locally Weighted Regression, and RBFs

[Guide](#) [Practice](#) [Sources](#)

Intuition

Nearby examples often have similar targets. kNN directly combines neighbour targets, LWR fits a small local regression around each query, and an RBF model turns distance from chosen centres into features.

Definitions and jargon

k is the neighbour count. A distance metric defines closeness. Bandwidth controls how quickly local influence decays. A radial basis function depends only on distance from its centre.

Formulas, symbols, shapes, and assumptions

Euclidean distance is $d(x,q)=\sqrt{\sum_j(x_j-q_j)^2}$. Weighted class vote can sum $1/d_i$ per class when every selected distance is positive; exact zero needs an explicit rule. LWR commonly uses $a_i=\exp(-\|x_i-q\|^2/(2\tau^2))$ and fits weighted least squares. An RBF feature is $\phi_j(x)=\exp(-\|x-c_j\|^2/(2s_j^2))$, with $\tau,s_j>0$.

Conditional properties

Small k or bandwidth makes a flexible, high-variance predictor; larger values smooth and may increase bias. Scaling matters because a large numeric feature range can dominate distance. In high dimensions, neighbours can become less distinctive as data grow sparse.

Important distinctions

kNN aggregates selected stored targets. LWR solves a query-specific weighted regression. An RBF model fits global coefficients on centre-based features. None is k-means clustering, although centres for an RBF model may be chosen using clustering.

Checked example

Three neighbours have labels A,A,B at distances 2,4,1. Uniform voting predicts A. Inverse-distance weights give A: $1/2+1/4=0.75$ and B: 1, so weighted voting predicts B.

Traps and counterexamples

Do not divide by zero for an identical neighbour. Do not choose k on the final test set. Euclidean distance is not meaningful until feature units and representation are considered.

Answer-free recall

- 1 Calculate uniform and weighted kNN predictions.
- 2 Explain the bandwidth bias-variance trade-off.
- 3 Distinguish kNN, LWR, RBF, and k-means.
- 4 State the zero-distance rule and explain why feature scaling can change neighbour selection.

Session 8 - Integrated Model Selection and Evaluation

Guide Practice Sources

Intuition

An end-to-end answer connects the task, data, leakage-safe fitting, selection, decision rule, and honest final estimate. Every formula needs its conditions.

Definitions and jargon

A hyperparameter, such as depth, k , bandwidth, or regularization, is chosen outside fitting. Generalization means useful behavior on new data. Calibration compares predicted probabilities with observed frequencies.

Formulas, symbols, shapes, and assumptions

Match output to method: a numerical target suggests regression; a category suggests classification. Match metric to harm: precision controls false-positive burden; recall controls missed positives; cost-sensitive risk directly weights decisions. Freeze preprocessing, features, model, hyperparameters, and threshold before final test evaluation.

Conditional properties

Random splits assume exchangeability and may fail across time or groups. Compare candidates on the same validation protocol. Tune thresholds on validation, not test.

Important distinctions

Model fitting is not model selection; model selection is not final testing. Better training loss need not mean better validation metric. A simple interpretable tree may be preferable to a slightly more accurate model when operational constraints require explanation.

Checked example

For a rare-fraud task, first split by time, fit preprocessing only on past training records, tune model and threshold on later validation records using expected cost or precision-recall, freeze the pipeline, and evaluate once on the newest test period. Randomly mixing future transactions into training would exaggerate deployment performance.

Traps and counterexamples

Repeatedly checking the test score turns the test set into a validation set. Accuracy can reward always predicting the majority. A probability threshold of 0.5 is a convention, not a universal optimum.

Answer-free recall

- 1 Design a leakage-safe workflow for a time-dependent task.
- 2 Choose a metric/model for stated costs; diagnose underfitting, overfitting, leakage, and threshold mismatch.